Predictive vs Generative

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected objective-and-evidence record.

Identify component targets, compute a distribution, follow the prefix and design useful evaluation checks.

Keep one habit: Retain counts, probabilities, conditional context, current observations and unverified claims separately.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Image and Voice Pipelines. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Image and Voice Pipelines

Feedback Loops

Similarity by Selected Traits

W1 · Identify each component’s target

> A combined application can classify, retrieve and generate.

Your evidence record

- Marker corpus for prefix The marker is: red6,blue3,green1,total10. The following table after red has end0.8,today0.2; it conditions on a different expanded prefix. The separate box corpus has full6,empty3,unknown1 with a current empty observation. Corpus counts do not establish a present physical state. W4 has its own signal counts and observation.

- Classify these components by target: label a message request/report; continue a prefix with a sequence; retrieve a stored sensor record. Explain why the application may combine them.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Compute the marker distribution

> Use the given corpus total and label its meaning.

Your evidence record

- Marker corpus for prefix The marker is: red6,blue3,green1,total10. The following table after red has end0.8,today0.2; it conditions on a different expanded prefix. The separate box corpus has full6,empty3,unknown1 with a current empty observation. Corpus counts do not establish a present physical state. W4 has its own signal counts and observation.

- For The marker is, calculate the three empirical probabilities, their sum and their meaning.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Follow the expanded prefix

> Different conditional tables describe different alternatives.

Your evidence record

- Marker corpus for prefix The marker is: red6,blue3,green1,total10. The following table after red has end0.8,today0.2; it conditions on a different expanded prefix. The separate box corpus has full6,empty3,unknown1 with a current empty observation. Corpus counts do not establish a present physical state. W4 has its own signal counts and observation.

- After red is selected, the expanded-prefix table gives end 0.8 and today 0.2. Explain why these values cannot be substituted for the red/blue/green probabilities.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Design distinct evidence and response checks

> Continuation fit alone cannot validate current-status claims.

Your evidence record

- Marker corpus for prefix The marker is: red6,blue3,green1,total10. The following table after red has end0.8,today0.2; it conditions on a different expanded prefix. The separate box corpus has full6,empty3,unknown1 with a current empty observation. Corpus counts do not establish a present physical state. W4 has its own signal counts and observation.

- A system reports current room status using retrieved measurements and generated sentences. Name checks for **evidence**, response quality and permission.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Analyse a new corpus independently

> Keep one first attempt before hints or feedback.

Read and explain independently

For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high.

State what the most frequent continuation does and does not establish.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Analyse a new corpus independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Keep scoring **predictive** and explanations templated or carefully generated and verified.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Compute the distribution for **Generative** and **predictive** objectives. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Generative** and **predictive** objectives. Proposed explanation: "**Generative** models do not predict". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A tool retrieves a book description, predicts a genre, then writes a new blurb. Identify each operation and give one distinct failure to check at each stage.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A labels genre; B writes a new blurb; C computes 24/6.

Distinguish operations.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

One model labels a review positive or negative; another writes a new review. Which classifies and which generates? Must generated text be true?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 27 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A tool retrieves a book description, predicts a genre, then writes a new blurb. Identify each operation and give one distinct failure to check at each stage.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high.

State what the most frequent continuation does and does not establish.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

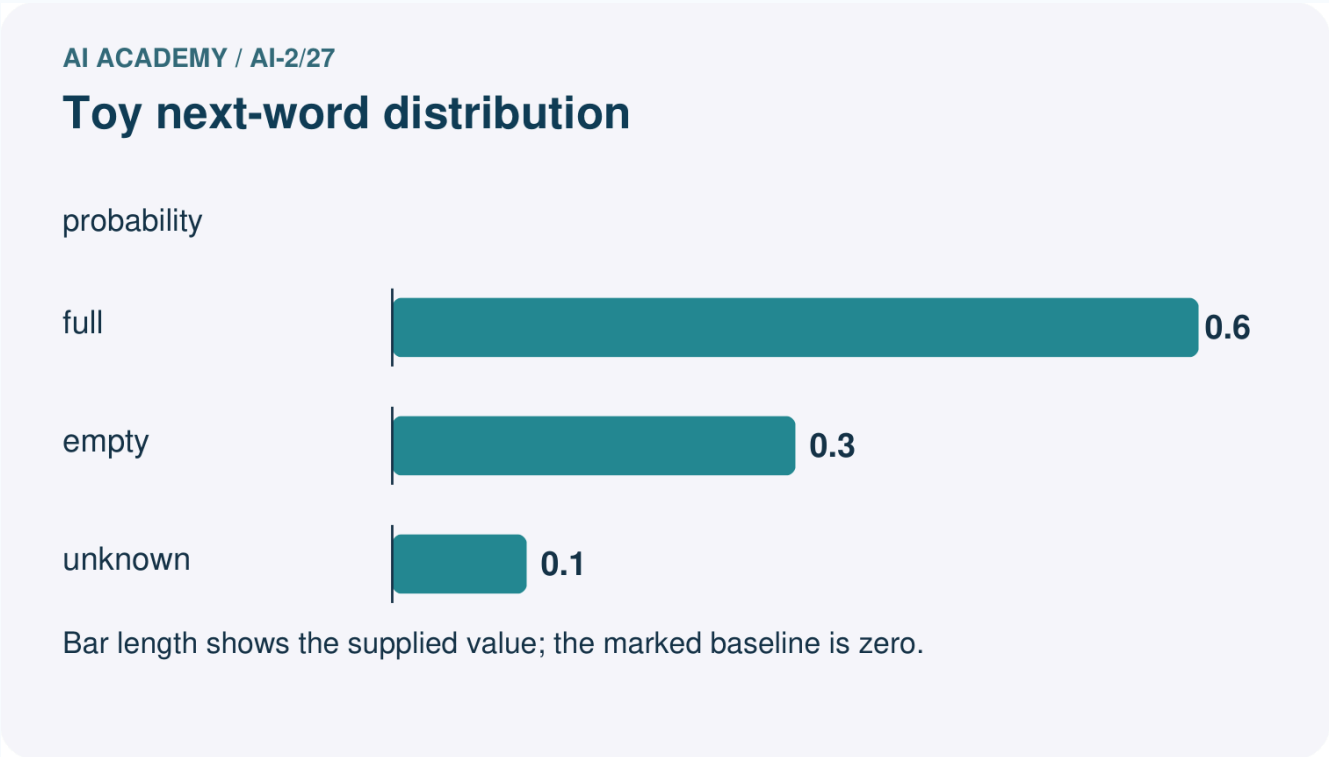
> Keep the explanation and the support you used.

Use this when helpful

A fictional toy generator predicts the next word after “The box is”.

Observed **training** continuations: full 6, empty 3, unknown 1.

Use counts divided by total 10. The current physical box is separately checked and found empty.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/27

Predictive vs Generative

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count table

Continuation counts are 6,3,1; total 10.



2. Probabilities

full=0.6; empty=0.3; unknown=0.1.



3. Greedy continuation

Largest probability chooses full for this toy prefix.



4. Check reality

Current observation says empty; frequent wording is not a present inventory reading.

CHANGE ONE THING

Change the training table to full 2, empty 7, unknown 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Does matching the physical box in the changed table prove the generator measures boxes?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Use approved book facts for recommender explanations; generated fluency is not a replacement for catalogue **evidence**.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use approved book facts for recommender explanations; generated fluency is not a replacement for catalogue **evidence**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional toy generator predicts the next word after "The box is".
- Which labelled step matches this detail? Current **observation** says empty; frequent wording is not a present inventory reading.
- 1. Probabilities
- 2. Count table
- 3. Greedy continuation
- 4. Check reality
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Check reality Current **observation** says empty; frequent wording is not a present inventory reading.

The table predicts a continuation in a text exercise, not the contents of the current box. Generating the most frequent token is a model-defined decision. A factual inventory statement needs current **evidence**.

The numerical table is intentionally small; real **generative** systems condition on richer context and do not simply consult these three counts.

Changed-case question

- Changed-case practice: Change the **training** table to full 2, empty 7, unknown 1. Which conclusion is supported by the supplied facts?
- 1. Greedy **output** becomes empty; the model assigns that token probability 0.7.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Greedy **output** becomes empty; the model assigns that token probability 0.7. The continuation distribution changed; the independently observed box has not changed.

Repair and save your work

Review Image and Voice Pipelines (ai-2/26). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Which stage first fails, and why should the faulty transcript not be silently overwritten?. Return to this week and explain what you changed.

Review Feedback Loops (ai-2/24). Goal: Implement a deterministic recommender. Short practice: What extra field would reveal why comparing raw clicks is misleading?. Return to this week and explain what you changed.

Review Similarity by Selected Traits (ai-2/21). Goal: Represent items with numeric features. Short practice: Explain why B ceases to be a closest option without any candidate data changing..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

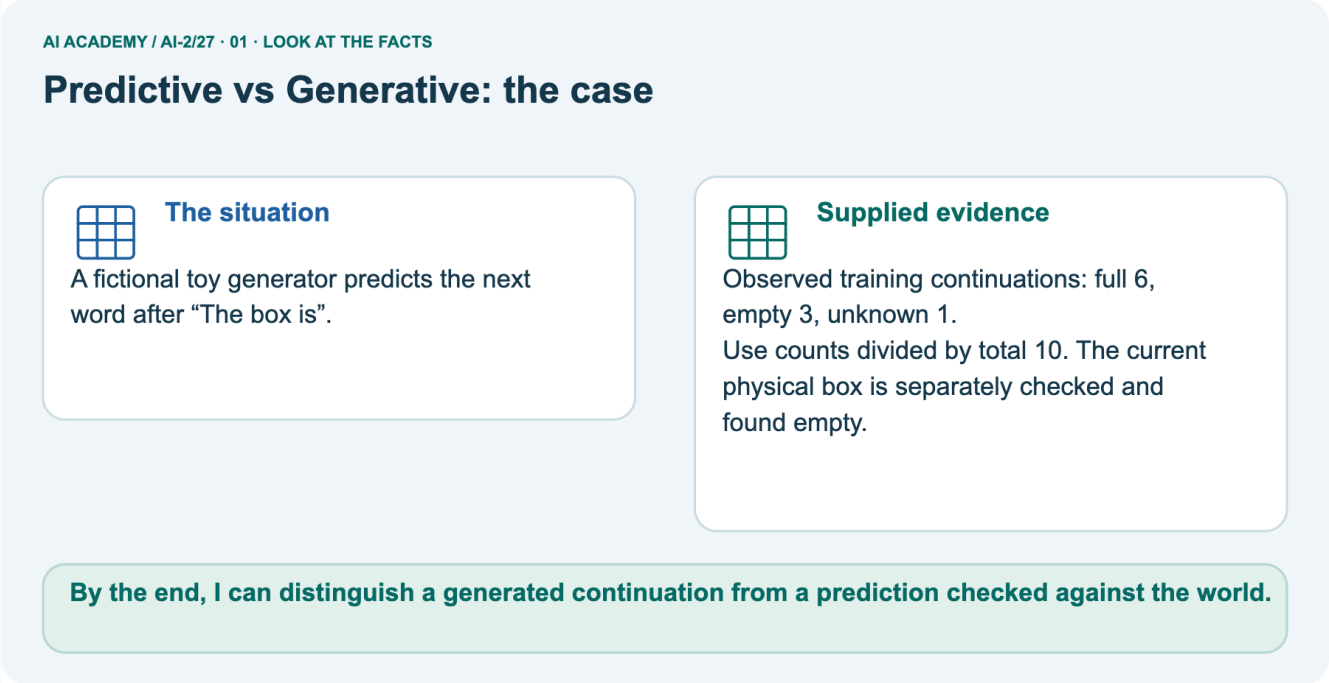
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Predictive vs Generative: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

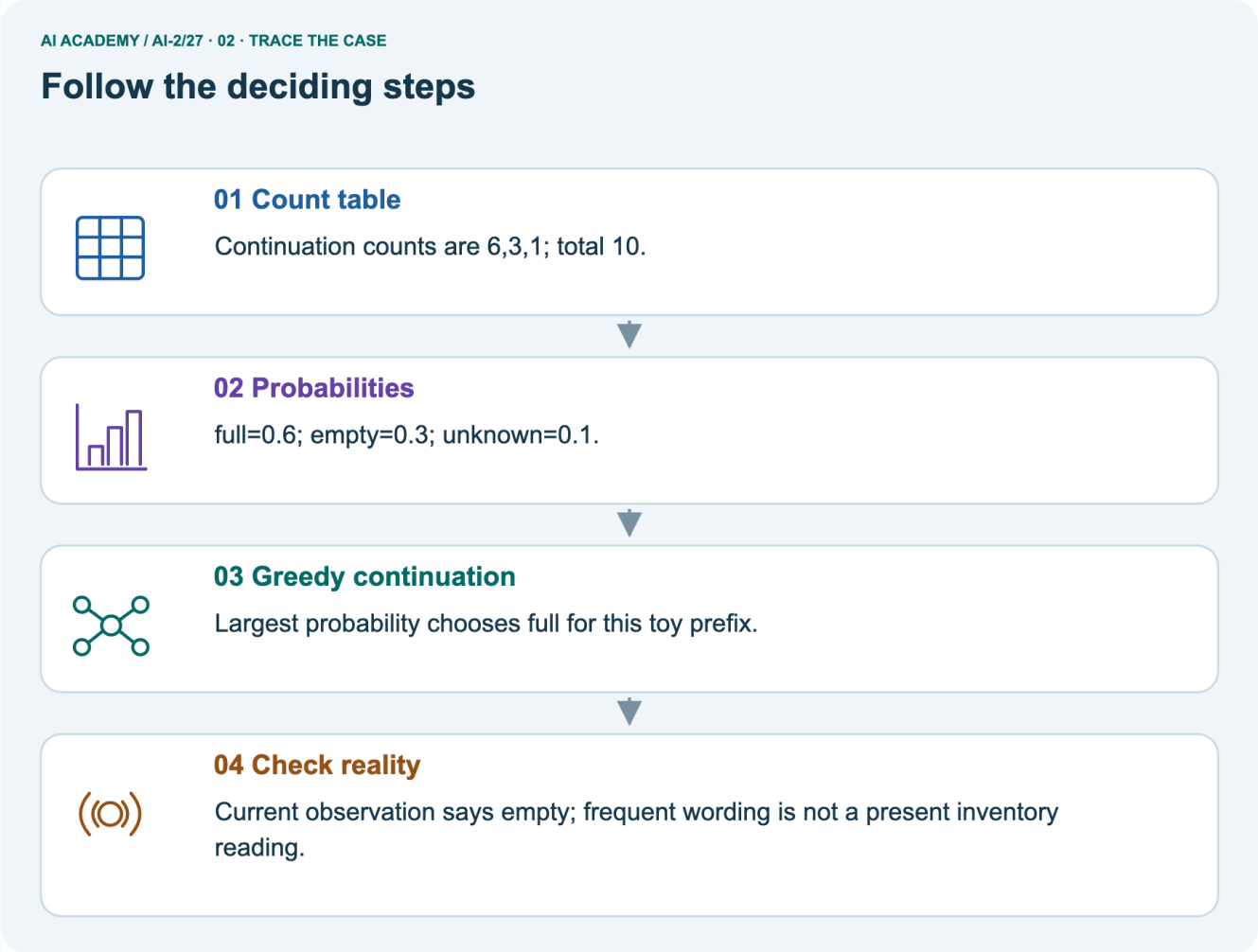
A fictional toy generator predicts the next word after “The box is”. Supplied facts: Observed **training** continuations: full 6, empty 3, unknown 1. Use counts divided by total 10.

The current physical box is separately checked and found empty.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



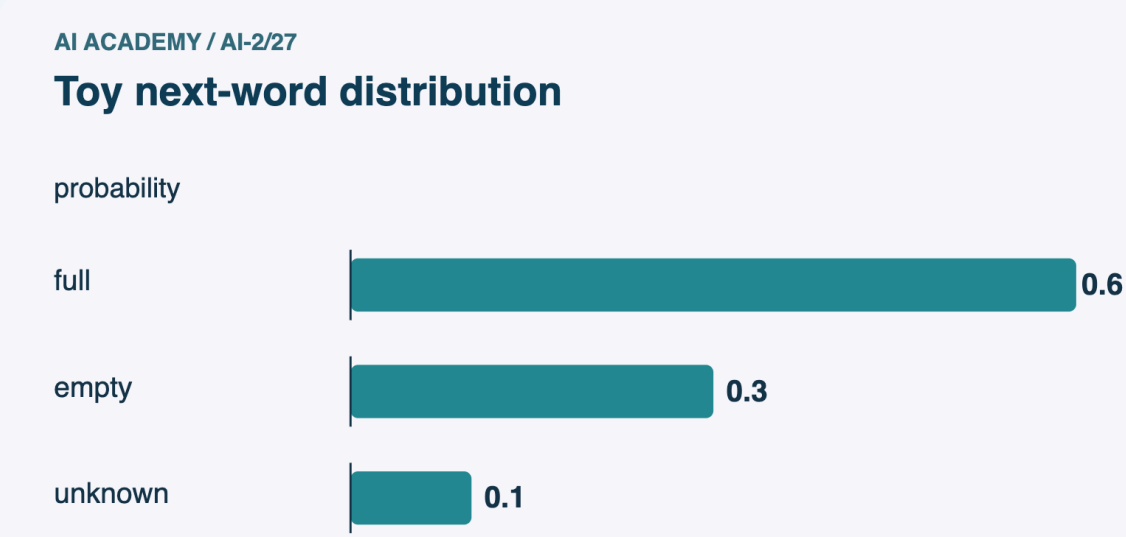
Follow the labelled picture with the matching explanation. Open the vector diagram

Count table: Continuation counts are 6,3,1; total 10. → Probabilities: full=0.6; empty=0.3; unknown=0.1. → Greedy continuation: Largest probability chooses full for this toy prefix. → Check reality: Current **observation** says empty; frequent wording is not a present inventory reading.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Toy next-word distribution



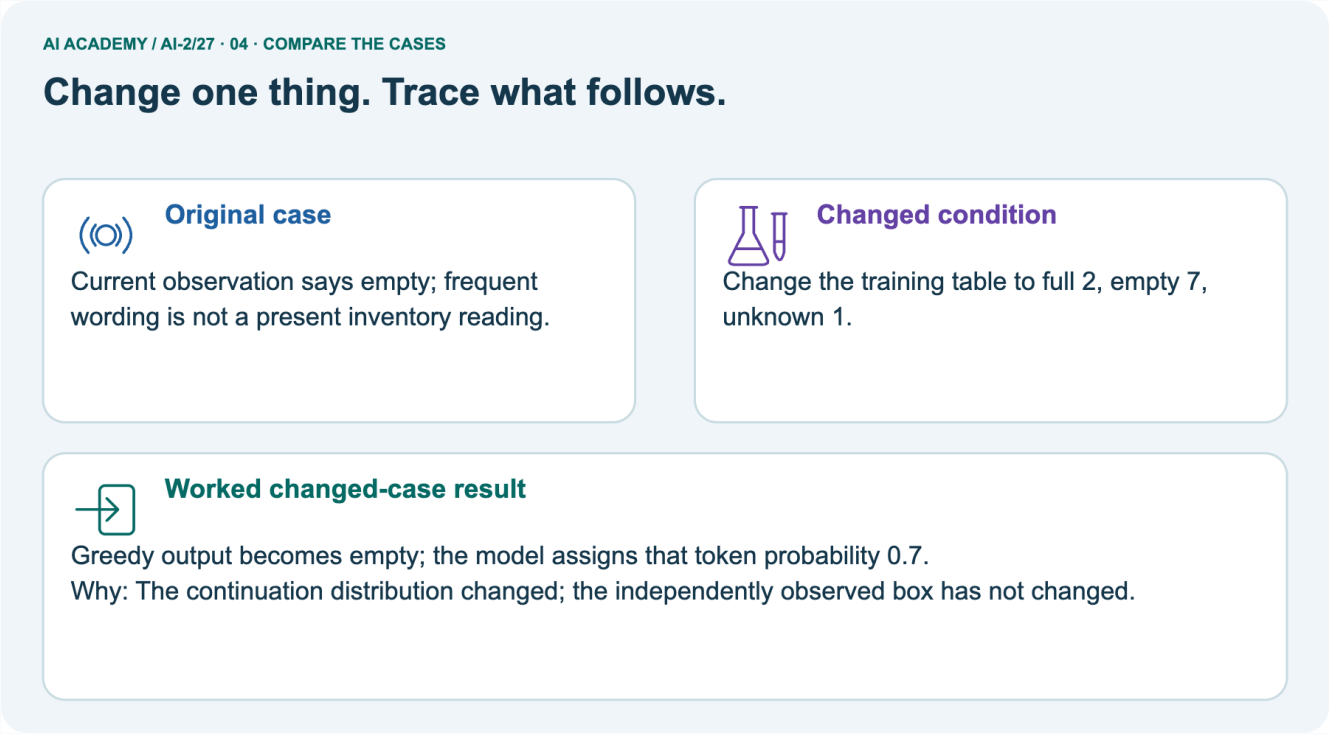
Follow the labelled picture with the matching explanation. Open the vector diagram

Toy next-word distribution. Units: probability. full: 0.6. empty: 0.3. unknown: 0.1.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

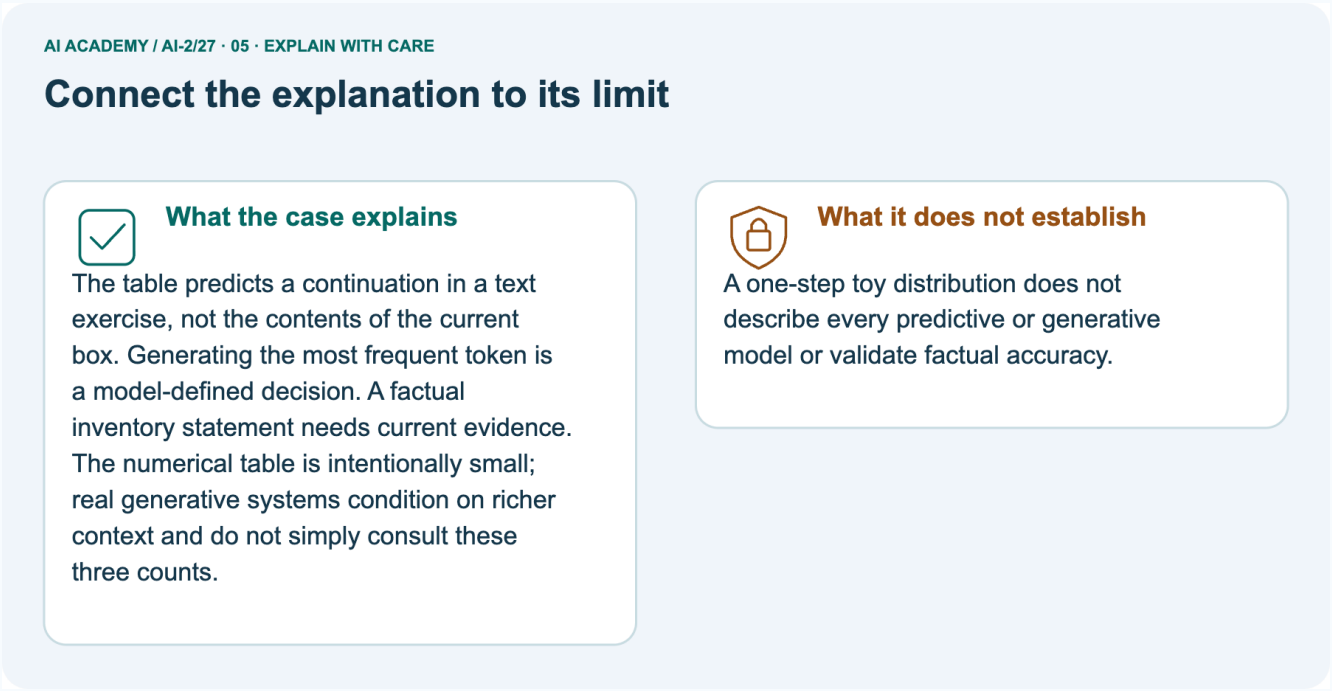
Original case: Current **observation** says empty; frequent wording is not a present inventory reading. Changed condition: Change the **training** table to full 2, empty 7, unknown 1.

Expected result: Greedy **output** becomes empty; the model assigns that token probability 0.7. Reason: The continuation distribution changed; the independently observed box has not changed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The table predicts a continuation in a text exercise, not the contents of the current box. Generating the most frequent token is a model-defined decision. A factual inventory statement needs current **evidence**.

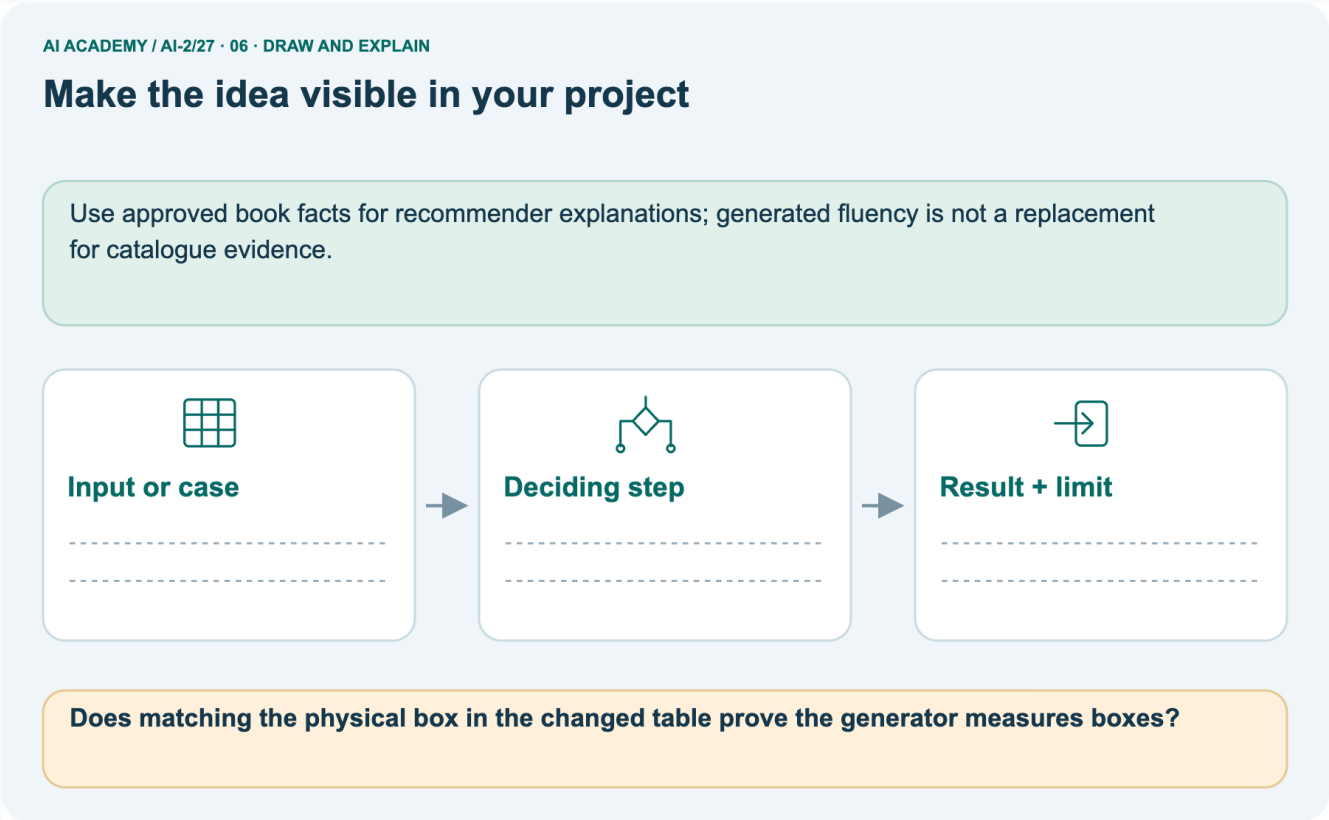
The numerical table is intentionally small; real **generative** systems condition on richer context and do not simply consult these three counts.

Limit: A one-step toy distribution does not describe every **predictive** or **generative** model or validate factual accuracy.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use approved book facts for recommender explanations; generated fluency is not a replacement for catalogue **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: Does matching the physical box in the changed table prove the generator measures boxes?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Predictive vs Generative: the case

02. Follow the deciding steps

03. Toy next-word distribution

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.